

Starting Comprehensive Sort Benchmark...

=====

Testing Size: 20

=====

-> Data Type: INT | Structure: RANDOM

- Quick Sort: 1.2384 seconds
- Merge Sort (Array): 1.8939 seconds
- Insertion Sort: 0.9908 seconds
- Bubble Sort: 1.6872 seconds
- Selection Sort: 1.1144 seconds
- Merge Sort (Linked List): 1.3183 seconds
- Stream Sort (Heap): 0.4349 seconds

-> Data Type: INT | Structure: SORTED

- Quick Sort: 0.8474 seconds
- Merge Sort (Array): 1.5001 seconds
- Insertion Sort: 0.2003 seconds
- Bubble Sort: 0.1192 seconds
- Selection Sort: 1.0272 seconds
- Merge Sort (Linked List): 1.1886 seconds
- Stream Sort (Heap): 0.3753 seconds

-> Data Type: INT | Structure: REVERSE

- Quick Sort: 0.8667 seconds
- Merge Sort (Array): 1.6519 seconds
- Insertion Sort: 1.5623 seconds

- Bubble Sort: 2.2482 seconds
- Selection Sort: 1.1264 seconds
- Merge Sort (Linked List): 1.1641 seconds
- Stream Sort (Heap): 0.3923 seconds

-> Data Type: INT | Structure: ALMOST

- Quick Sort: 0.9278 seconds
- Merge Sort (Array): 1.5613 seconds
- Insertion Sort: 0.2845 seconds
- Bubble Sort: 0.6933 seconds
- Selection Sort: 1.0400 seconds
- Merge Sort (Linked List): 1.2232 seconds
- Stream Sort (Heap): 0.3849 seconds

-> Data Type: INT | Structure: MIXED

- Quick Sort: 1.0033 seconds
- Merge Sort (Array): 1.6045 seconds
- Insertion Sort: 0.3583 seconds
- Bubble Sort: 0.7907 seconds
- Selection Sort: 1.0502 seconds
- Merge Sort (Linked List): 1.2736 seconds
- Stream Sort (Heap): 0.3873 seconds

-> Data Type: FLOAT | Structure: RANDOM

- Quick Sort: 1.3700 seconds
- Merge Sort (Array): 1.7528 seconds
- Insertion Sort: 0.9151 seconds
- Bubble Sort: 1.6366 seconds

- Selection Sort: 1.0196 seconds
- Merge Sort (Linked List): 1.3765 seconds
- Stream Sort (Heap): 0.4297 seconds

-> Data Type: FLOAT | Structure: SORTED

- Quick Sort: 1.0331 seconds
- Merge Sort (Array): 1.4240 seconds
- Insertion Sort: 0.1834 seconds
- Bubble Sort: 0.1121 seconds
- Selection Sort: 0.9727 seconds
- Merge Sort (Linked List): 1.1963 seconds
- Stream Sort (Heap): 0.3830 seconds

-> Data Type: FLOAT | Structure: REVERSE

- Quick Sort: 1.0275 seconds
- Merge Sort (Array): 1.5584 seconds
- Insertion Sort: 1.6502 seconds
- Bubble Sort: 2.2562 seconds
- Selection Sort: 1.0457 seconds
- Merge Sort (Linked List): 1.1755 seconds
- Stream Sort (Heap): 0.4005 seconds

-> Data Type: FLOAT | Structure: ALMOST

- Quick Sort: 1.0655 seconds
- Merge Sort (Array): 1.5152 seconds
- Insertion Sort: 0.2762 seconds
- Bubble Sort: 0.6758 seconds
- Selection Sort: 0.9849 seconds

- Merge Sort (Linked List): 1.2407 seconds
- Stream Sort (Heap): 0.4015 seconds

-> Data Type: FLOAT | Structure: MIXED

- Quick Sort: 1.1738 seconds
- Merge Sort (Array): 1.5466 seconds
- Insertion Sort: 0.3926 seconds
- Bubble Sort: 0.7811 seconds
- Selection Sort: 0.9985 seconds
- Merge Sort (Linked List): 1.2807 seconds
- Stream Sort (Heap): 0.4590 seconds

-> Data Type: STRING | Structure: RANDOM

- Quick Sort: 1.5477 seconds
- Merge Sort (Array): 1.8590 seconds
- Insertion Sort: 1.0524 seconds
- Bubble Sort: 1.9446 seconds
- Selection Sort: 1.2952 seconds
- Merge Sort (Linked List): 1.4496 seconds
- Stream Sort (Heap): 0.4470 seconds

-> Data Type: STRING | Structure: SORTED

- Quick Sort: 1.2208 seconds
- Merge Sort (Array): 1.4989 seconds
- Insertion Sort: 0.2136 seconds
- Bubble Sort: 0.1476 seconds
- Selection Sort: 1.2388 seconds
- Merge Sort (Linked List): 1.2708 seconds

- Stream Sort (Heap): 0.4069 seconds

-> Data Type: STRING | Structure: REVERSE

- Quick Sort: 1.3581 seconds
- Merge Sort (Array): 1.6525 seconds
- Insertion Sort: 1.8322 seconds
- Bubble Sort: 2.5355 seconds
- Selection Sort: 1.3345 seconds
- Merge Sort (Linked List): 1.2288 seconds
- Stream Sort (Heap): 0.4262 seconds

-> Data Type: STRING | Structure: ALMOST

- Quick Sort: 1.2407 seconds
- Merge Sort (Array): 1.6126 seconds
- Insertion Sort: 0.3314 seconds
- Bubble Sort: 0.8319 seconds
- Selection Sort: 1.3338 seconds
- Merge Sort (Linked List): 1.3057 seconds
- Stream Sort (Heap): 0.4360 seconds

-> Data Type: STRING | Structure: MIXED

- Quick Sort: 1.3787 seconds
- Merge Sort (Array): 1.7830 seconds
- Insertion Sort: 0.4417 seconds
- Bubble Sort: 0.9721 seconds
- Selection Sort: 1.2790 seconds
- Merge Sort (Linked List): 1.3097 seconds
- Stream Sort (Heap): 0.4771 seconds

=====

Testing Size: 30

=====

-> Data Type: INT | Structure: RANDOM

- Quick Sort: 1.9489 seconds
- Merge Sort (Array): 2.8318 seconds
- Insertion Sort: 1.9255 seconds
- Bubble Sort: 3.6918 seconds
- Selection Sort: 2.1753 seconds
- Merge Sort (Linked List): 2.3004 seconds
- Stream Sort (Heap): 0.6771 seconds

-> Data Type: INT | Structure: SORTED

- Quick Sort: 1.3930 seconds
- Merge Sort (Array): 2.3609 seconds
- Insertion Sort: 0.2819 seconds
- Bubble Sort: 0.1646 seconds
- Selection Sort: 2.0883 seconds
- Merge Sort (Linked List): 1.8944 seconds
- Stream Sort (Heap): 0.4802 seconds

-> Data Type: INT | Structure: REVERSE

- Quick Sort: 1.3999 seconds
- Merge Sort (Array): 2.4109 seconds
- Insertion Sort: 3.5220 seconds
- Bubble Sort: 5.2521 seconds

- Selection Sort: 2.2664 seconds
- Merge Sort (Linked List): 1.9473 seconds
- Stream Sort (Heap): 0.5086 seconds

-> Data Type: INT | Structure: ALMOST

- Quick Sort: 1.5905 seconds
- Merge Sort (Array): 2.4961 seconds
- Insertion Sort: 0.4292 seconds
- Bubble Sort: 1.3968 seconds
- Selection Sort: 2.1000 seconds
- Merge Sort (Linked List): 1.9559 seconds
- Stream Sort (Heap): 0.5003 seconds

-> Data Type: INT | Structure: MIXED

- Quick Sort: 1.7773 seconds
- Merge Sort (Array): 2.5859 seconds
- Insertion Sort: 0.6862 seconds
- Bubble Sort: 1.7706 seconds
- Selection Sort: 2.1289 seconds
- Merge Sort (Linked List): 1.9259 seconds
- Stream Sort (Heap): 0.5330 seconds

-> Data Type: FLOAT | Structure: RANDOM

- Quick Sort: 2.3576 seconds
- Merge Sort (Array): 2.9089 seconds
- Insertion Sort: 1.8949 seconds
- Bubble Sort: 3.5789 seconds
- Selection Sort: 2.0525 seconds

- Merge Sort (Linked List): 2.2858 seconds
- Stream Sort (Heap): 0.5949 seconds

-> Data Type: FLOAT | Structure: SORTED

- Quick Sort: 1.5308 seconds
- Merge Sort (Array): 2.2974 seconds
- Insertion Sort: 0.2737 seconds
- Bubble Sort: 0.1577 seconds
- Selection Sort: 1.9424 seconds
- Merge Sort (Linked List): 1.8887 seconds
- Stream Sort (Heap): 0.4954 seconds

-> Data Type: FLOAT | Structure: REVERSE

- Quick Sort: 1.4788 seconds
- Merge Sort (Array): 2.4147 seconds
- Insertion Sort: 3.5931 seconds
- Bubble Sort: 4.8866 seconds
- Selection Sort: 2.0977 seconds
- Merge Sort (Linked List): 1.9164 seconds
- Stream Sort (Heap): 0.5232 seconds

-> Data Type: FLOAT | Structure: ALMOST

- Quick Sort: 1.6640 seconds
- Merge Sort (Array): 2.4048 seconds
- Insertion Sort: 0.4193 seconds
- Bubble Sort: 1.3199 seconds
- Selection Sort: 1.9560 seconds
- Merge Sort (Linked List): 2.0498 seconds

- Stream Sort (Heap): 0.6294 seconds

-> Data Type: FLOAT | Structure: MIXED

- Quick Sort: 1.9992 seconds
- Merge Sort (Array): 2.4771 seconds
- Insertion Sort: 0.7071 seconds
- Bubble Sort: 1.6919 seconds
- Selection Sort: 2.0059 seconds
- Merge Sort (Linked List): 1.9697 seconds
- Stream Sort (Heap): 0.5539 seconds

-> Data Type: STRING | Structure: RANDOM

- Quick Sort: 2.6286 seconds
- Merge Sort (Array): 2.9943 seconds
- Insertion Sort: 2.3531 seconds
- Bubble Sort: 4.2125 seconds
- Selection Sort: 2.6679 seconds
- Merge Sort (Linked List): 2.4235 seconds
- Stream Sort (Heap): 0.6116 seconds

-> Data Type: STRING | Structure: SORTED

- Quick Sort: 1.8191 seconds
- Merge Sort (Array): 2.4296 seconds
- Insertion Sort: 0.3313 seconds
- Bubble Sort: 0.2155 seconds
- Selection Sort: 2.5726 seconds
- Merge Sort (Linked List): 2.0885 seconds
- Stream Sort (Heap): 0.5447 seconds

-> Data Type: STRING | Structure: REVERSE

- Quick Sort: 1.8392 seconds
- Merge Sort (Array): 2.5584 seconds
- Insertion Sort: 4.0688 seconds
- Bubble Sort: 5.5553 seconds
- Selection Sort: 2.7760 seconds
- Merge Sort (Linked List): 2.0447 seconds
- Stream Sort (Heap): 0.5732 seconds

-> Data Type: STRING | Structure: ALMOST

- Quick Sort: 1.9681 seconds
- Merge Sort (Array): 2.5195 seconds
- Insertion Sort: 0.5048 seconds
- Bubble Sort: 1.7034 seconds
- Selection Sort: 2.5702 seconds
- Merge Sort (Linked List): 2.0835 seconds
- Stream Sort (Heap): 0.5623 seconds

-> Data Type: STRING | Structure: MIXED

- Quick Sort: 2.2941 seconds
- Merge Sort (Array): 2.9081 seconds
- Insertion Sort: 0.8400 seconds
- Bubble Sort: 2.1705 seconds
- Selection Sort: 2.6228 seconds
- Merge Sort (Linked List): 2.0775 seconds
- Stream Sort (Heap): 0.6157 seconds

=====

Testing Size: 50

=====

-> Data Type: INT | Structure: RANDOM

- Quick Sort: 3.6189 seconds
- Merge Sort (Array): 5.1916 seconds
- Insertion Sort: 5.3186 seconds
- Bubble Sort: 10.0515 seconds
- Selection Sort: 5.4785 seconds
- Merge Sort (Linked List): 4.3313 seconds
- Stream Sort (Heap): 0.9083 seconds

-> Data Type: INT | Structure: SORTED

- Quick Sort: 2.6509 seconds
- Merge Sort (Array): 4.1790 seconds
- Insertion Sort: 0.4676 seconds
- Bubble Sort: 0.2661 seconds
- Selection Sort: 5.2752 seconds
- Merge Sort (Linked List): 3.6412 seconds
- Stream Sort (Heap): 0.7259 seconds

-> Data Type: INT | Structure: REVERSE

- Quick Sort: 2.6752 seconds
- Merge Sort (Array): 4.3700 seconds
- Insertion Sort: 9.6877 seconds
- Bubble Sort: 13.7350 seconds
- Selection Sort: 5.7146 seconds

- Merge Sort (Linked List): 3.5432 seconds
- Stream Sort (Heap): 0.7593 seconds

-> Data Type: INT | Structure: ALMOST

- Quick Sort: 2.7118 seconds
- Merge Sort (Array): 4.4704 seconds
- Insertion Sort: 0.7184 seconds
- Bubble Sort: 3.4384 seconds
- Selection Sort: 5.2611 seconds
- Merge Sort (Linked List): 3.5556 seconds
- Stream Sort (Heap): 0.7491 seconds

-> Data Type: INT | Structure: MIXED

- Quick Sort: 3.1404 seconds
- Merge Sort (Array): 4.6409 seconds
- Insertion Sort: 1.5959 seconds
- Bubble Sort: 4.9721 seconds
- Selection Sort: 5.3128 seconds
- Merge Sort (Linked List): 3.6489 seconds
- Stream Sort (Heap): 0.7990 seconds

-> Data Type: FLOAT | Structure: RANDOM

- Quick Sort: 4.2760 seconds
- Merge Sort (Array): 5.1951 seconds
- Insertion Sort: 5.0072 seconds
- Bubble Sort: 9.5770 seconds
- Selection Sort: 5.2940 seconds
- Merge Sort (Linked List): 4.2409 seconds

- Stream Sort (Heap): 0.9271 seconds

-> Data Type: FLOAT | Structure: SORTED

- Quick Sort: 3.1101 seconds
- Merge Sort (Array): 3.9938 seconds
- Insertion Sort: 0.4551 seconds
- Bubble Sort: 0.2499 seconds
- Selection Sort: 4.8569 seconds
- Merge Sort (Linked List): 3.5063 seconds
- Stream Sort (Heap): 0.7391 seconds

-> Data Type: FLOAT | Structure: REVERSE

- Quick Sort: 3.0808 seconds
- Merge Sort (Array): 4.3518 seconds
- Insertion Sort: 9.5034 seconds
- Bubble Sort: 13.4770 seconds
- Selection Sort: 5.3454 seconds
- Merge Sort (Linked List): 3.6281 seconds
- Stream Sort (Heap): 0.7793 seconds

-> Data Type: FLOAT | Structure: ALMOST

- Quick Sort: 3.1975 seconds
- Merge Sort (Array): 4.2303 seconds
- Insertion Sort: 0.7158 seconds
- Bubble Sort: 3.1912 seconds
- Selection Sort: 5.0854 seconds
- Merge Sort (Linked List): 3.6092 seconds
- Stream Sort (Heap): 0.9179 seconds

-> Data Type: FLOAT | Structure: MIXED

- Quick Sort: 3.6515 seconds
- Merge Sort (Array): 4.4792 seconds
- Insertion Sort: 1.6253 seconds
- Bubble Sort: 4.6694 seconds
- Selection Sort: 5.0212 seconds
- Merge Sort (Linked List): 3.8245 seconds
- Stream Sort (Heap): 0.8711 seconds

-> Data Type: STRING | Structure: RANDOM

- Quick Sort: 4.8970 seconds
- Merge Sort (Array): 5.5174 seconds
- Insertion Sort: 5.9703 seconds
- Bubble Sort: 11.4930 seconds
- Selection Sort: 6.7932 seconds
- Merge Sort (Linked List): 4.6855 seconds
- Stream Sort (Heap): 0.9671 seconds

-> Data Type: STRING | Structure: SORTED

- Quick Sort: 3.7659 seconds
- Merge Sort (Array): 4.2997 seconds
- Insertion Sort: 0.5802 seconds
- Bubble Sort: 0.3702 seconds
- Selection Sort: 6.9384 seconds
- Merge Sort (Linked List): 3.7916 seconds
- Stream Sort (Heap): 0.8469 seconds

-> Data Type: STRING | Structure: REVERSE

- Quick Sort: 3.7680 seconds
- Merge Sort (Array): 4.6242 seconds
- Insertion Sort: 11.2743 seconds
- Bubble Sort: 15.1892 seconds
- Selection Sort: 7.3509 seconds
- Merge Sort (Linked List): 3.7033 seconds
- Stream Sort (Heap): 0.8590 seconds

-> Data Type: STRING | Structure: ALMOST

- Quick Sort: 3.8786 seconds
- Merge Sort (Array): 4.4789 seconds
- Insertion Sort: 0.8693 seconds
- Bubble Sort: 4.4912 seconds
- Selection Sort: 6.8605 seconds
- Merge Sort (Linked List): 3.8595 seconds
- Stream Sort (Heap): 0.9130 seconds

-> Data Type: STRING | Structure: MIXED

- Quick Sort: 4.3168 seconds
- Merge Sort (Array): 4.9802 seconds
- Insertion Sort: 1.9533 seconds
- Bubble Sort: 6.0281 seconds
- Selection Sort: 6.7576 seconds
- Merge Sort (Linked List): 4.1349 seconds
- Stream Sort (Heap): 0.9693 seconds

=====

Testing Size: 100

=====

-> Data Type: INT | Structure: RANDOM

- Quick Sort: 8.3768 seconds
- Merge Sort (Array): 11.8386 seconds
- Insertion Sort: 20.0328 seconds
- Bubble Sort: 40.2012 seconds
- Selection Sort: 20.0344 seconds
- Merge Sort (Linked List): 9.5134 seconds
- Stream Sort (Heap): 1.8075 seconds

-> Data Type: INT | Structure: SORTED

- Quick Sort: 5.9878 seconds
- Merge Sort (Array): 9.2084 seconds
- Insertion Sort: 0.9505 seconds
- Bubble Sort: 0.5113 seconds
- Selection Sort: 19.6858 seconds
- Merge Sort (Linked List): 7.9685 seconds
- Stream Sort (Heap): 1.3416 seconds

-> Data Type: INT | Structure: REVERSE

- Quick Sort: 6.0581 seconds
- Merge Sort (Array): 9.3551 seconds
- Insertion Sort: 38.6635 seconds
- Bubble Sort: 53.5267 seconds
- Selection Sort: 21.4480 seconds
- Merge Sort (Linked List): 7.9982 seconds

- Stream Sort (Heap): 1.3886 seconds

-> Data Type: INT | Structure: ALMOST

- Quick Sort: 6.1537 seconds
- Merge Sort (Array): 9.7282 seconds
- Insertion Sort: 2.0241 seconds
- Bubble Sort: 17.5137 seconds
- Selection Sort: 19.6608 seconds
- Merge Sort (Linked List): 8.0974 seconds
- Stream Sort (Heap): 1.4243 seconds

-> Data Type: INT | Structure: MIXED

- Quick Sort: 7.3455 seconds
- Merge Sort (Array): 10.1377 seconds
- Insertion Sort: 5.6084 seconds
- Bubble Sort: 20.0866 seconds
- Selection Sort: 19.8688 seconds
- Merge Sort (Linked List): 8.4759 seconds
- Stream Sort (Heap): 1.5362 seconds

-> Data Type: FLOAT | Structure: RANDOM

- Quick Sort: 9.5846 seconds
- Merge Sort (Array): 11.9842 seconds
- Insertion Sort: 19.1151 seconds
- Bubble Sort: 37.9882 seconds
- Selection Sort: 18.5878 seconds
- Merge Sort (Linked List): 9.7432 seconds
- Stream Sort (Heap): 1.8857 seconds

-> Data Type: FLOAT | Structure: SORTED

- Quick Sort: 6.9921 seconds
- Merge Sort (Array): 9.1647 seconds
- Insertion Sort: 0.9708 seconds
- Bubble Sort: 0.5331 seconds
- Selection Sort: 17.9901 seconds
- Merge Sort (Linked List): 7.9970 seconds
- Stream Sort (Heap): 1.4622 seconds

-> Data Type: FLOAT | Structure: REVERSE

- Quick Sort: 6.8880 seconds
- Merge Sort (Array): 9.4218 seconds
- Insertion Sort: 37.3752 seconds
- Bubble Sort: 51.8145 seconds
- Selection Sort: 19.7838 seconds
- Merge Sort (Linked List): 7.7601 seconds
- Stream Sort (Heap): 1.5318 seconds

-> Data Type: FLOAT | Structure: ALMOST

- Quick Sort: 7.1880 seconds
- Merge Sort (Array): 9.4519 seconds
- Insertion Sort: 1.9823 seconds
- Bubble Sort: 16.0002 seconds
- Selection Sort: 18.2059 seconds
- Merge Sort (Linked List): 8.2316 seconds
- Stream Sort (Heap): 1.6123 seconds

-> Data Type: FLOAT | Structure: MIXED

- Quick Sort: 8.4302 seconds
- Merge Sort (Array): 9.9500 seconds
- Insertion Sort: 5.5566 seconds
- Bubble Sort: 18.7810 seconds
- Selection Sort: 18.6983 seconds
- Merge Sort (Linked List): 8.6346 seconds
- Stream Sort (Heap): 1.7033 seconds

-> Data Type: STRING | Structure: RANDOM

- Quick Sort: 11.5657 seconds
- Merge Sort (Array): 12.6492 seconds
- Insertion Sort: 22.7754 seconds
- Bubble Sort: 45.7516 seconds
- Selection Sort: 25.5191 seconds
- Merge Sort (Linked List): 10.4648 seconds
- Stream Sort (Heap): 1.9901 seconds

-> Data Type: STRING | Structure: SORTED

- Quick Sort: 8.6198 seconds
- Merge Sort (Array): 9.6060 seconds
- Insertion Sort: 1.2253 seconds
- Bubble Sort: 0.7787 seconds
- Selection Sort: 25.4498 seconds
- Merge Sort (Linked List): 8.4216 seconds
- Stream Sort (Heap): 1.5808 seconds

-> Data Type: STRING | Structure: REVERSE

- Quick Sort: 8.9542 seconds
- Merge Sort (Array): 9.9661 seconds
- Insertion Sort: 44.2306 seconds
- Bubble Sort: 60.0707 seconds
- Selection Sort: 27.3781 seconds
- Merge Sort (Linked List): 8.4277 seconds
- Stream Sort (Heap): 1.7329 seconds

-> Data Type: STRING | Structure: ALMOST

- Quick Sort: 8.8715 seconds
- Merge Sort (Array): 10.1124 seconds
- Insertion Sort: 2.5194 seconds
- Bubble Sort: 21.9082 seconds
- Selection Sort: 25.3562 seconds
- Merge Sort (Linked List): 9.0623 seconds
- Stream Sort (Heap): 1.8150 seconds

-> Data Type: STRING | Structure: MIXED

- Quick Sort: 10.2095 seconds
- Merge Sort (Array): 10.8231 seconds
- Insertion Sort: 6.6608 seconds
- Bubble Sort: 24.5787 seconds
- Selection Sort: 25.6566 seconds
- Merge Sort (Linked List): 9.1841 seconds
- Stream Sort (Heap): 1.9440 seconds

=====

Testing Size: 1000

=====

-> Data Type: INT | Structure: RANDOM

- Quick Sort: 0.0015 seconds
- Merge Sort (Array): 0.0018 seconds
- Insertion Sort: 0.0217 seconds
- Bubble Sort: 0.0467 seconds
- Selection Sort: 0.0241 seconds
- Merge Sort (Linked List): 0.0017 seconds
- Stream Sort (Heap): 0.0011 seconds

-> Data Type: INT | Structure: SORTED

- Quick Sort: 0.0013 seconds
- Merge Sort (Array): 0.0019 seconds
- Insertion Sort: 0.0001 seconds
- Bubble Sort: 0.0001 seconds
- Selection Sort: 0.0215 seconds
- Merge Sort (Linked List): 0.0011 seconds
- Stream Sort (Heap): 0.0002 seconds

-> Data Type: INT | Structure: REVERSE

- Quick Sort: 0.0009 seconds
- Merge Sort (Array): 0.0026 seconds
- Insertion Sort: 0.0442 seconds
- Bubble Sort: 0.0617 seconds
- Selection Sort: 0.0253 seconds
- Merge Sort (Linked List): 0.0013 seconds
- Stream Sort (Heap): 0.0003 seconds

-> Data Type: INT | Structure: ALMOST

- Quick Sort: 0.0010 seconds
- Merge Sort (Array): 0.0019 seconds
- Insertion Sort: 0.0010 seconds
- Bubble Sort: 0.0383 seconds
- Selection Sort: 0.0300 seconds
- Merge Sort (Linked List): 0.0016 seconds
- Stream Sort (Heap): 0.0002 seconds

-> Data Type: INT | Structure: MIXED

- Quick Sort: 0.0011 seconds
- Merge Sort (Array): 0.0027 seconds
- Insertion Sort: 0.0098 seconds
- Bubble Sort: 0.0245 seconds
- Selection Sort: 0.0214 seconds
- Merge Sort (Linked List): 0.0023 seconds
- Stream Sort (Heap): 0.0003 seconds

-> Data Type: FLOAT | Structure: RANDOM

- Quick Sort: 0.0016 seconds
- Merge Sort (Array): 0.0026 seconds
- Insertion Sort: 0.0247 seconds
- Bubble Sort: 0.0525 seconds
- Selection Sort: 0.0244 seconds
- Merge Sort (Linked List): 0.0022 seconds
- Stream Sort (Heap): 0.0006 seconds

-> Data Type: FLOAT | Structure: SORTED

- Quick Sort: 0.0009 seconds
- Merge Sort (Array): 0.0011 seconds
- Insertion Sort: 0.0001 seconds
- Bubble Sort: 0.0001 seconds
- Selection Sort: 0.0215 seconds
- Merge Sort (Linked List): 0.0012 seconds
- Stream Sort (Heap): 0.0002 seconds

-> Data Type: FLOAT | Structure: REVERSE

- Quick Sort: 0.0009 seconds
- Merge Sort (Array): 0.0011 seconds
- Insertion Sort: 0.0430 seconds
- Bubble Sort: 0.0595 seconds
- Selection Sort: 0.0230 seconds
- Merge Sort (Linked List): 0.0014 seconds
- Stream Sort (Heap): 0.0003 seconds

-> Data Type: FLOAT | Structure: ALMOST

- Quick Sort: 0.0011 seconds
- Merge Sort (Array): 0.0015 seconds
- Insertion Sort: 0.0011 seconds
- Bubble Sort: 0.0260 seconds
- Selection Sort: 0.0209 seconds
- Merge Sort (Linked List): 0.0015 seconds
- Stream Sort (Heap): 0.0002 seconds

-> Data Type: FLOAT | Structure: MIXED

- Quick Sort: 0.0011 seconds
- Merge Sort (Array): 0.0014 seconds
- Insertion Sort: 0.0063 seconds
- Bubble Sort: 0.0235 seconds
- Selection Sort: 0.0222 seconds
- Merge Sort (Linked List): 0.0015 seconds
- Stream Sort (Heap): 0.0003 seconds

-> Data Type: STRING | Structure: RANDOM

- Quick Sort: 0.0017 seconds
- Merge Sort (Array): 0.0018 seconds
- Insertion Sort: 0.0247 seconds
- Bubble Sort: 0.0523 seconds
- Selection Sort: 0.0261 seconds
- Merge Sort (Linked List): 0.0021 seconds
- Stream Sort (Heap): 0.0003 seconds

-> Data Type: STRING | Structure: SORTED

- Quick Sort: 0.0014 seconds
- Merge Sort (Array): 0.0019 seconds
- Insertion Sort: 0.0001 seconds
- Bubble Sort: 0.0001 seconds
- Selection Sort: 0.0282 seconds
- Merge Sort (Linked List): 0.0020 seconds
- Stream Sort (Heap): 0.0002 seconds

-> Data Type: STRING | Structure: REVERSE

- Quick Sort: 0.0015 seconds

- Merge Sort (Array): 0.0015 seconds
- Insertion Sort: 0.0499 seconds
- Bubble Sort: 0.0661 seconds
- Selection Sort: 0.0277 seconds
- Merge Sort (Linked List): 0.0017 seconds
- Stream Sort (Heap): 0.0002 seconds

-> Data Type: STRING | Structure: ALMOST

- Quick Sort: 0.0012 seconds
- Merge Sort (Array): 0.0015 seconds
- Insertion Sort: 0.0019 seconds
- Bubble Sort: 0.0314 seconds
- Selection Sort: 0.0275 seconds
- Merge Sort (Linked List): 0.0024 seconds
- Stream Sort (Heap): 0.0002 seconds

-> Data Type: STRING | Structure: MIXED

- Quick Sort: 0.0016 seconds
- Merge Sort (Array): 0.0015 seconds
- Insertion Sort: 0.0070 seconds
- Bubble Sort: 0.0314 seconds
- Selection Sort: 0.0274 seconds
- Merge Sort (Linked List): 0.0022 seconds
- Stream Sort (Heap): 0.0006 seconds

=====

Testing Size: 10000

=====

-> Data Type: INT | Structure: RANDOM

- Quick Sort: 0.0177 seconds
- Merge Sort (Array): 0.0249 seconds
- Insertion Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Bubble Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Selection Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Merge Sort (Linked List): 0.0270 seconds
- Parallel Merge Sort: 0.4039 seconds
- Stream Sort (Heap): 0.0041 seconds

-> Data Type: INT | Structure: SORTED

- Quick Sort: 0.0114 seconds
- Merge Sort (Array): 0.0156 seconds
- Insertion Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Bubble Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Selection Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Merge Sort (Linked List): 0.0151 seconds
- Parallel Merge Sort: 0.1376 seconds
- Stream Sort (Heap): 0.0019 seconds

-> Data Type: INT | Structure: REVERSE

- Quick Sort: 0.0111 seconds
- Merge Sort (Array): 0.0176 seconds
- Insertion Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Bubble Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Selection Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Merge Sort (Linked List): 0.0147 seconds

- Parallel Merge Sort: 0.1527 seconds
- Stream Sort (Heap): 0.0018 seconds

-> Data Type: INT | Structure: ALMOST

- Quick Sort: 0.0115 seconds
- Merge Sort (Array): 0.0191 seconds
- Insertion Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Bubble Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Selection Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Merge Sort (Linked List): 0.0207 seconds
- Parallel Merge Sort: 0.1429 seconds
- Stream Sort (Heap): 0.0020 seconds

-> Data Type: INT | Structure: MIXED

- Quick Sort: 0.0144 seconds
- Merge Sort (Array): 0.0177 seconds
- Insertion Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Bubble Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Selection Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Merge Sort (Linked List): 0.0185 seconds
- Parallel Merge Sort: 0.1583 seconds
- Stream Sort (Heap): 0.0038 seconds

-> Data Type: FLOAT | Structure: RANDOM

- Quick Sort: 0.0178 seconds
- Merge Sort (Array): 0.0214 seconds
- Insertion Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Bubble Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]

- Selection Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Merge Sort (Linked List): 0.0229 seconds
- Parallel Merge Sort: 0.1633 seconds
- Stream Sort (Heap): 0.0035 seconds

-> Data Type: FLOAT | Structure: SORTED

- Quick Sort: 0.0121 seconds
- Merge Sort (Array): 0.0146 seconds
- Insertion Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Bubble Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Selection Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Merge Sort (Linked List): 0.0150 seconds
- Parallel Merge Sort: 0.1327 seconds
- Stream Sort (Heap): 0.0018 seconds

-> Data Type: FLOAT | Structure: REVERSE

- Quick Sort: 0.0113 seconds
- Merge Sort (Array): 0.0166 seconds
- Insertion Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Bubble Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Selection Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Merge Sort (Linked List): 0.0172 seconds
- Parallel Merge Sort: 0.1861 seconds
- Stream Sort (Heap): 0.0018 seconds

-> Data Type: FLOAT | Structure: ALMOST

- Quick Sort: 0.0119 seconds
- Merge Sort (Array): 0.0196 seconds

- Insertion Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Bubble Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Selection Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Merge Sort (Linked List): 0.0203 seconds
- Parallel Merge Sort: 0.1464 seconds
- Stream Sort (Heap): 0.0033 seconds

-> Data Type: FLOAT | Structure: MIXED

- Quick Sort: 0.0146 seconds
- Merge Sort (Array): 0.0197 seconds
- Insertion Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Bubble Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Selection Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Merge Sort (Linked List): 0.0189 seconds
- Parallel Merge Sort: 0.2539 seconds
- Stream Sort (Heap): 0.0028 seconds

-> Data Type: STRING | Structure: RANDOM

- Quick Sort: 0.0222 seconds
- Merge Sort (Array): 0.0233 seconds
- Insertion Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Bubble Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Selection Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Merge Sort (Linked List): 0.0270 seconds
- Parallel Merge Sort: 0.1438 seconds
- Stream Sort (Heap): 0.0045 seconds

-> Data Type: STRING | Structure: SORTED

- Quick Sort: 0.0151 seconds
- Merge Sort (Array): 0.0149 seconds
- Insertion Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Bubble Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Selection Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Merge Sort (Linked List): 0.0163 seconds
- Parallel Merge Sort: 0.1323 seconds
- Stream Sort (Heap): 0.0022 seconds

-> Data Type: STRING | Structure: REVERSE

- Quick Sort: 0.0153 seconds
- Merge Sort (Array): 0.0178 seconds
- Insertion Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Bubble Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Selection Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Merge Sort (Linked List): 0.0154 seconds
- Parallel Merge Sort: 0.1363 seconds
- Stream Sort (Heap): 0.0019 seconds

-> Data Type: STRING | Structure: ALMOST

- Quick Sort: 0.0160 seconds
- Merge Sort (Array): 0.0199 seconds
- Insertion Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Bubble Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Selection Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Merge Sort (Linked List): 0.0233 seconds
- Parallel Merge Sort: 0.1788 seconds
- Stream Sort (Heap): 0.0028 seconds

-> Data Type: STRING | Structure: MIXED

- Quick Sort: 0.0196 seconds
- Merge Sort (Array): 0.0183 seconds
- Insertion Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Bubble Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Selection Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Merge Sort (Linked List): 0.0192 seconds
- Parallel Merge Sort: 0.1348 seconds
- Stream Sort (Heap): 0.0027 seconds

=====

Testing Size: 1000000

=====

-> Data Type: INT | Structure: RANDOM

- Quick Sort: 3.1630 seconds
- Merge Sort (Array): 3.5560 seconds
- Insertion Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Bubble Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Selection Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Merge Sort (Linked List): [SKIPPED - Size too large for Python recursion]
- Parallel Merge Sort: 0.9191 seconds
- Stream Sort (Heap): 0.6426 seconds

-> Data Type: INT | Structure: SORTED

- Quick Sort: 2.5940 seconds
- Merge Sort (Array): 2.4453 seconds

- Insertion Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Bubble Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Selection Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Merge Sort (Linked List): [SKIPPED - Size too large for Python recursion]
- Parallel Merge Sort: 0.7489 seconds
- Stream Sort (Heap): 0.6493 seconds

-> Data Type: INT | Structure: REVERSE

- Quick Sort: 2.7652 seconds
- Merge Sort (Array): 2.3138 seconds
- Insertion Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Bubble Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Selection Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Merge Sort (Linked List): [SKIPPED - Size too large for Python recursion]
- Parallel Merge Sort: 0.7241 seconds
- Stream Sort (Heap): 0.6298 seconds

-> Data Type: INT | Structure: ALMOST

- Quick Sort: 2.6150 seconds
- Merge Sort (Array): 3.5565 seconds
- Insertion Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Bubble Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Selection Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Merge Sort (Linked List): [SKIPPED - Size too large for Python recursion]
- Parallel Merge Sort: 0.9986 seconds
- Stream Sort (Heap): 0.6948 seconds

-> Data Type: INT | Structure: MIXED

- Quick Sort: 3.3813 seconds
- Merge Sort (Array): 3.1112 seconds
- Insertion Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Bubble Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Selection Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Merge Sort (Linked List): [SKIPPED - Size too large for Python recursion]
- Parallel Merge Sort: 0.8680 seconds
- Stream Sort (Heap): 0.7127 seconds

-> Data Type: INT | Structure: FLAT

- Quick Sort: 0.1951 seconds
- Merge Sort (Array): 2.8123 seconds
- Insertion Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Bubble Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Selection Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Merge Sort (Linked List): [SKIPPED - Size too large for Python recursion]
- Parallel Merge Sort: 0.8501 seconds
- Stream Sort (Heap): 0.5352 seconds

-> Data Type: FLOAT | Structure: RANDOM

- Quick Sort: 3.3745 seconds
- Merge Sort (Array): 3.5380 seconds
- Insertion Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Bubble Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Selection Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Merge Sort (Linked List): [SKIPPED - Size too large for Python recursion]
- Parallel Merge Sort: 0.9788 seconds
- Stream Sort (Heap): 0.6838 seconds

-> Data Type: FLOAT | Structure: SORTED

- Quick Sort: 2.5981 seconds
- Merge Sort (Array): 2.3664 seconds
- Insertion Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Bubble Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Selection Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Merge Sort (Linked List): [SKIPPED - Size too large for Python recursion]
- Parallel Merge Sort: 0.8110 seconds
- Stream Sort (Heap): 0.6425 seconds

-> Data Type: FLOAT | Structure: REVERSE

- Quick Sort: 2.7590 seconds
- Merge Sort (Array): 2.3338 seconds
- Insertion Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Bubble Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Selection Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Merge Sort (Linked List): [SKIPPED - Size too large for Python recursion]
- Parallel Merge Sort: 0.7917 seconds
- Stream Sort (Heap): 0.6746 seconds

-> Data Type: FLOAT | Structure: ALMOST

- Quick Sort: 2.6579 seconds
- Merge Sort (Array): 3.6853 seconds
- Insertion Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Bubble Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Selection Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Merge Sort (Linked List): [SKIPPED - Size too large for Python recursion]

- Parallel Merge Sort: 1.0512 seconds
- Stream Sort (Heap): 0.7337 seconds

-> Data Type: FLOAT | Structure: MIXED

- Quick Sort: 3.5163 seconds
- Merge Sort (Array): 3.1783 seconds
- Insertion Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Bubble Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Selection Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Merge Sort (Linked List): [SKIPPED - Size too large for Python recursion]
- Parallel Merge Sort: 0.9087 seconds
- Stream Sort (Heap): 0.7309 seconds

-> Data Type: FLOAT | Structure: FLAT

- Quick Sort: 0.1981 seconds
- Merge Sort (Array): 2.7695 seconds
- Insertion Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Bubble Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Selection Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Merge Sort (Linked List): [SKIPPED - Size too large for Python recursion]
- Parallel Merge Sort: 0.8786 seconds
- Stream Sort (Heap): 0.5223 seconds

-> Data Type: STRING | Structure: RANDOM

- Quick Sort: 4.3655 seconds
- Merge Sort (Array): 3.7912 seconds
- Insertion Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Bubble Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]

- Selection Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Merge Sort (Linked List): [SKIPPED - Size too large for Python recursion]
- Parallel Merge Sort: 1.0665 seconds
- Stream Sort (Heap): 0.8386 seconds

-> Data Type: STRING | Structure: SORTED

- Quick Sort: 4.0890 seconds
- Merge Sort (Array): 2.6034 seconds
- Insertion Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Bubble Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Selection Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Merge Sort (Linked List): [SKIPPED - Size too large for Python recursion]
- Parallel Merge Sort: 0.8980 seconds
- Stream Sort (Heap): 0.7392 seconds

-> Data Type: STRING | Structure: REVERSE

- Quick Sort: 4.1281 seconds
- Merge Sort (Array): 2.4296 seconds
- Insertion Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Bubble Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Selection Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Merge Sort (Linked List): [SKIPPED - Size too large for Python recursion]
- Parallel Merge Sort: 0.9071 seconds
- Stream Sort (Heap): 0.7560 seconds

-> Data Type: STRING | Structure: ALMOST

- Quick Sort: 3.9507 seconds
- Merge Sort (Array): 3.7230 seconds

- Insertion Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Bubble Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Selection Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Merge Sort (Linked List): [SKIPPED - Size too large for Python recursion]
- Parallel Merge Sort: 1.2100 seconds
- Stream Sort (Heap): 0.8819 seconds

-> Data Type: STRING | Structure: MIXED

- Quick Sort: 5.3474 seconds
- Merge Sort (Array): 3.2779 seconds
- Insertion Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Bubble Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Selection Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Merge Sort (Linked List): [SKIPPED - Size too large for Python recursion]
- Parallel Merge Sort: 1.0384 seconds
- Stream Sort (Heap): 0.8614 seconds

-> Data Type: STRING | Structure: FLAT

- Quick Sort: 0.2419 seconds
- Merge Sort (Array): 2.8602 seconds
- Insertion Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Bubble Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Selection Sort: [SKIPPED - $O(n^2)$ too slow for size > 1000]
- Merge Sort (Linked List): [SKIPPED - Size too large for Python recursion]
- Parallel Merge Sort: 0.8207 seconds
- Stream Sort (Heap): 0.5283 seconds